

A Dynamic and Modular Architecture for Language Modeling

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Abstract

Large language models based on transformer architectures have demonstrated impressive generative capabilities, but remain monolithic systems with fixed context windows, implicit reasoning, and no explicit cognitive structure. Their performance depends primarily on scale, and they lack mechanisms for domain isolation, epistemic evaluation, or deterministic control flow.

This paper introduces Dynamic Modular Language Models (DMLMs), a fundamentally different architecture for language modeling. A DMLM consists of many small, domain-bound micro-models that interact through explicit interfaces, local paging structures, and plural evaluators governing grammar, semantics, coherence, and world logic. A PRE loop (Proposal–Retry–Evaluate) orchestrates recursion, stabilization, and long-form generation.

We show that DMLMs produce coherent narrative structure, stable long-range dependencies, and emergent world models using compact curricula and without transformers or large-scale training. The architecture provides interpretability, modular extensibility, and deterministic reasoning, positioning DMLMs as a distinct class of language models: dynamic, modular, and epistemically governed rather than monolithic and scale-driven.

Contents

1	Introduction	3
1.1	Contributions of This Paper	3
2	Motivation	3
2.1	Why This Paper Exists	4
3	Related Work	4
3.1	Society of Mind and Agentic Cognition	4
3.2	Cognitive Architectures	5
3.3	Multi-Agent and Tool-Augmented LLM Systems	5
3.4	Modular and Mixture-of-Experts Models	5
3.5	Modular and Configurable Language Models	6
3.6	Composable Expert-Based Inference Frameworks	6
3.7	Large Language Models and Transformer Architectures	6
3.8	How a DMLM Differs Fundamentally	7

4	Architecture	7
4.1	Micro-Models and Writer Agents	7
4.2	Paging as Emergent Micro-Memory	7
4.3	Dynamic and Incremental Learning	8
4.4	Transition Maps as Learned Continuity	8
4.5	Evaluators and Epistemic Governance	8
4.6	The PRE Loop: Proposal–Retry–Evaluate	9
4.7	Canonical Form: Internal vs. External Language	9
4.8	Curriculum as Language Definition	9
4.9	Learned Attention and Context Compression	9
5	Relation to the Paged-PRE-CAS AGI Architecture	10
5.1	Paged Reasoning at the Level of Writer Agents	10
5.2	PRE: From Proposal–Evaluate–Refine to Proposal–Evaluate–Retry	10
5.3	Optional Token-Level PRE for Efficiency	11
5.4	Optional CAS Vectors for Epistemic Guidance	11
5.5	M-Paged-PRE: A Multi-Agent Reasoning Ensemble	11
5.6	Conceptual Alignment	12
6	Example Output	12
7	Limitations and Open Questions	12
7.1	Limitations	12
7.2	Open Questions	13
8	Testable Predictions	13
8.1	Prediction 1: Stable Long-Range Coherence Through Paging	13
8.2	Prediction 2: Evaluator-Governed Convergence	14
8.3	Prediction 3: Domain-Bound Behavior of Micro-Models	14
8.4	Prediction 4: Emergent Structure Without Scale	14
8.5	Prediction 5: Reproducible Behavior Under Fixed Configuration	14
8.6	Prediction 6: Modular Extensibility Without Retraining	14
8.7	Prediction 7: PRE Loop Produces Distinct Regimes	14
8.8	Prediction 8: Local World Models Are Persistent	14
8.9	Prediction 9: Generation Efficiency Improves with Better Evaluators and Curricula	15
9	Conclusion	15
A	Triadic Cosmos Ecosystem Context	16

1 Introduction

Language models have advanced rapidly in recent years, driven primarily by the scaling of transformer architectures. These models demonstrate impressive fluency and broad task coverage, yet their underlying structure remains monolithic: a single global parameter space, a fixed context window, and no explicit cognitive organization. Their reasoning is implicit, their memory is transient, and their behavior is governed by statistical correlations rather than modular computation.

This paper presents a different approach. Dynamic Modular Language Models (DMLMs) introduce an architectural alternative to monolithic transformers by decomposing language generation into interacting micro-models governed by explicit evaluators, paging structures, and a deterministic PRE loop. Instead of relying on scale, DMLMs rely on structure: domain-bound modules, local memory, explicit interfaces, and epistemic governance.

The full theoretical and conceptual foundations of this architecture are developed in the *Triadic Cosmos* ecosystem, which provides a broader ontological and computational framework for modular intelligence. The present paper does not attempt to reproduce that material. Instead, it offers a concise architectural overview of DMLMs, focusing on the core mechanisms required to understand, implement, and evaluate the model. Readers interested in the deeper theoretical context, extended examples, and the full architectural philosophy are encouraged to consult the accompanying book and ecosystem documentation.

1.1 Contributions of This Paper

This paper makes the following contributions:

- **Architectural formulation.** We present a compact, self-contained description of Dynamic Modular Language Models, including paging, evaluators, Writer Agents, and the PRE loop.
- **Structural motivation.** We identify the architectural limitations of monolithic transformer-based LLMs and explain why modularity, explicit evaluators, and local memory are necessary for stable long-form generation.
- **Operational clarity.** We provide a clear account of how DMLMs function in practice: how micro-models interact, how evaluators govern reasoning, and how paging structures maintain coherence.
- **Empirical illustration.** We summarize key empirical observations from the broader ecosystem—emergent narrative structure, stable recursion, and domain-bound world models—without reproducing the full experimental corpus.
- **Pointer to the ecosystem.** We situate this paper within the larger *Triadic Cosmos* framework and direct readers to the book and repositories for the complete theoretical, architectural, and empirical development.

In short, this paper serves as a concise architectural entry point into Dynamic Modular Language Models: a modular, evaluator-governed alternative to monolithic language models, designed to make the core ideas accessible while inviting readers to explore the full ecosystem.

2 Motivation

Modern language models have achieved remarkable fluency, coherence, and task performance, but their underlying architecture has remained fundamentally unchanged since the introduction

of the transformer. These models are monolithic, scale-driven systems: a single global parameter space, a fixed context window, and no explicit cognitive structure. Their reasoning is implicit, their memory is transient, and their behavior is governed by statistical correlations rather than modular computation.

As a result, current LLMs face structural limitations that cannot be resolved by further scaling alone. They lack domain boundaries, epistemic evaluators, and deterministic control flow. They cannot maintain stable long-range world models, cannot reason with explicit internal state, and cannot decompose tasks into interacting cognitive units. Their architecture provides no mechanism for modular extension, interpretability, or principled governance.

DMLMs are motivated by the need for an alternative: a language model architecture that is not monolithic but modular, not scale-driven but structure-driven, not implicitly cognitive but explicitly organized. Instead of relying on a single transformer, a DMLM consists of many small, domain-bound micro-models that interact through paging, evaluators, and a PRE loop that governs recursion and stabilization.

2.1 Why This Paper Exists

This paper exists to articulate and formalize a fundamentally different approach to language modeling. While prior work has explored modularity within neural networks, multi-agent LLM systems, and cognitive architectures, none provide a unified, operational framework for structured, evaluator-governed, modular language generation.

The goal of this paper is to:

- identify the structural limitations of monolithic transformer-based LLMs,
- introduce a modular architecture that provides explicit cognitive structure,
- demonstrate how paging, evaluators, and micro-models enable stable long-form generation,
- show that emergent linguistic and narrative structure can arise without large-scale training,
- and position DMLMs as a distinct class of language models with architectural, not statistical, foundations.

In short, this paper exists because the field needs an alternative to monolithic LLMs—an architecture that is modular, interpretable, extensible, and epistemically governed. DMLMs offer such an alternative, providing a path toward structured, stable, and transparent language generation without reliance on massive datasets or transformer-scale computation.

3 Related Work

Research on cognitive architectures, modular AI systems, and language models spans several decades. Dynamic Modular Language Models (DMLMs) draw from multiple traditions—symbolic architectures, agent-based cognition, modular neural systems, and multi-agent LLM frameworks—while differing fundamentally from all of them in structure, execution model, and epistemic governance.

3.1 Society of Mind and Agentic Cognition

Minsky’s *Society of Mind* Minsky 1986 proposed that intelligence emerges from the interaction of many small, specialized agents rather than from a single monolithic process. This perspective anticipated modern views of distributed cognition, modular intelligence, and multi-agent reasoning.

However, the agents in Society of Mind lack explicit interfaces, epistemic evaluators, domain boundaries, and deterministic control flow. The framework does not specify how agents should coordinate, how conflicts should be resolved, or how reasoning should be governed at scale. It provides a conceptual metaphor rather than an operational architecture.

DMLMs can be viewed as a structural and computational realization of the Society of Mind intuition: they implement explicit agent roles (Writer Agents), deterministic evaluators, domain-bound paging structures, and a PRE loop that governs interaction and recursion. Where Society of Mind offers a philosophical lens, DMLMs provide a concrete, reproducible, and empirically testable architecture.

3.2 Cognitive Architectures

Classical cognitive architectures such as Soar Laird 2012, ACT-R Anderson and Lebiere 1998, CLARION Sun 2002, and Global Workspace Theory Baars 1988; Dehaene 2014 model cognition as structured interactions between symbolic components, memory systems, and control processes. These frameworks emphasize modularity, explicit reasoning, and interpretable internal state.

However, they rely on hand-crafted rules, lack scalable learning mechanisms, and do not provide a path toward domain expansion or cloud-native execution. Their modularity is symbolic rather than computational, and their learning mechanisms do not generalize to open-ended domains.

DMLMs inherit their emphasis on structure and transparency while introducing scalable recursion, plural evaluators, distributed paging, and domain-bound micro-models that learn from compact curricula. Unlike classical architectures, DMLMs are fully data-driven, dynamically extensible, and capable of emergent behavior without hand-coded rules.

3.3 Multi-Agent and Tool-Augmented LLM Systems

Recent frameworks such as AutoGen Research 2023, LangChain LangChain 2023, and Microsoft’s Semantic Kernel Microsoft 2023 orchestrate large language models with external tools, memory stores, and multi-agent workflows. These systems demonstrate practical utility and highlight the benefits of decomposing tasks across multiple interacting components.

However, their reasoning is emergent rather than explicit, their state is stored in unstructured transcripts, and their planning relies on prompting rather than algorithmic control. They lack domain isolation, deterministic execution, and epistemic governance. Their agents do not possess internal structure, memory, or evaluators; instead, they rely on the underlying LLM to simulate these capabilities.

In contrast, DMLMs implement reasoning as a first-class computational process, independent of language models. Writer Agents have explicit paging structures, local memory, and deterministic evaluators. The PRE loop governs interaction, recursion, and stabilization, providing a principled alternative to prompt-driven multi-agent LLM systems.

3.4 Modular and Mixture-of-Experts Models

Mixture-of-experts architectures Shazeer et al. 2017; Lepikhin et al. 2021; Fedus, Zoph, and Shazeer 2022 introduce modularity at the level of neural computation, enabling specialization and improved efficiency. While these models demonstrate the benefits of distributed computation, they remain monolithic in their parameterization and do not provide explicit cognitive roles, domain boundaries, or traceable reasoning steps.

Their modularity is statistical rather than structural: experts are selected by learned routing mechanisms, but the model remains a single global function with shared training dynamics. There is no notion of local memory, domain-specific evaluators, or multi-agent negotiation.

DMLMs extend modularity to the cognitive level. Each micro-model operates within a local paging structure, interacts through explicit interfaces, and is governed by deterministic evaluators. Modularity is architectural rather than statistical, enabling interpretability, traceability, and emergent structure without monolithic parameterization.

3.5 Modular and Configurable Language Models

Recent work has begun to explore modularity *within* large language models themselves. Configurable and modular LMs decompose a pretrained transformer into functional components or “bricks”, enabling partial activation, specialization, or reconfiguration at inference time Xiao et al. 2023; Qiu et al. 2024; Shen et al. 2024. These approaches highlight that modular structure can emerge inside otherwise monolithic architectures and can be exploited for efficiency, continual learning, or interpretability.

However, these models remain fundamentally tied to a single global transformer backbone. Their modules are internal partitions of a shared parameter space rather than independent, domain-bound cognitive units with explicit interfaces, local memory, and evaluators. In contrast, DMLMs treat modules as separate micro-models embedded in a paging and evaluation architecture, decoupling cognitive structure from any particular neural substrate.

3.6 Composable Expert-Based Inference Frameworks

Composable inference frameworks decompose queries into subproblems and route them to specialized experts or models, often organized as graphs or pipelines. These systems improve efficiency and task coverage by composing multiple experts at inference time, but they typically operate over otherwise monolithic LLMs and lack persistent cognitive structure, local world models, or epistemic evaluators.

DMLMs share the intuition of expert composition but differ in that the composition is not an external orchestration layer: it is built into the architecture itself via paging, evaluators, and the PRE loop, yielding a stable, recurrent cognitive substrate rather than an ad-hoc routing mechanism.

3.7 Large Language Models and Transformer Architectures

Transformer-based language models Vaswani et al. 2017 established the dominant architecture for modern language modeling. GPT-2 Radford et al. 2019 demonstrated that scaling transformer decoders leads to coherent long-form text generation, contextual continuity, and emergent linguistic structure. Its release marked a turning point in the field: a single monolithic model trained on a large corpus could generate surprisingly fluent text without task-specific supervision.

However, GPT-2 also revealed the structural limitations of monolithic transformer models. Its context window is fixed, its memory is transient, and its reasoning is implicit rather than explicit. The model lacks domain boundaries, epistemic evaluators, and deterministic control flow. Scaling improves fluency but does not introduce cognitive structure.

These limitations motivated the development of GPT-3 Brown et al. 2020, which showed that few-shot learning and broader generalization emerge at larger scales. Neural scaling laws Kaplan et al. 2020 further formalized the relationship between model size, dataset size, and performance.

DMLMs address the structural limitations of both GPT-2 and GPT-3 by introducing modularity at the architectural level rather than the statistical level. Paging, evaluators, and domain-bound micro-models provide explicit structure, local memory, and epistemic governance, enabling coherent long-form generation without reliance on massive datasets or monolithic transformers.

3.8 How a DMLM Differs Fundamentally

DMLMs differ from prior work in four fundamental ways:

1. **Architectural Modularity.** DMLMs consist of many small, domain-bound micro-models rather than a single global network. Their interactions are governed by explicit interfaces and evaluators.
2. **Deterministic Epistemic Governance.** Evaluators enforce grammar, semantics, coherence, and world physics, providing stability and preventing drift—capabilities absent in both classical architectures and modern LLM systems.
3. **Paging as Cognitive Structure.** Paging creates local memory, local attractors, and local world models, enabling scalable long-form generation without global context windows.
4. **Emergence Without Scale.** DMLMs produce grammar, roles, worlds, and narrative structure from compact curricula, without transformers, massive datasets, or monolithic training.

Together, these properties position DMLMs as a distinct class of language models: neither symbolic nor neural, neither monolithic nor prompt-driven, but a dynamic, modular, and interpretable architecture for language generation.

4 Architecture

Dynamic Modular Language Models (DMLMs) decompose language generation into many small, interacting components—Writer Agents, Token Pages, transition maps, evaluators, and a deterministic PRE loop. Unlike monolithic transformer architectures, which are static snapshots of a single training run, DMLMs are dynamic systems: they grow, reorganize, and refine their internal structure throughout their lifetime. This section provides a high-level overview of the architecture; full technical details appear in the companion Triadic Cosmos volume.

4.1 Micro-Models and Writer Agents

A DMLM consists of multiple *Writer Agents*, each a small, self-contained micro-LM with:

- its own Token Pages and transition maps,
- its own miniature prediction networks,
- its own curriculum and training trajectory,
- its own lemma clusters, attention patterns, and emergent world.

Agents do not share parameters. They diverge naturally due to differences in paging structure, transition growth, and interaction history. A central *Writer Engine* orchestrates these agents, selecting which one proposes the next sentence and switching agents when retries fail. This multi-agent orchestration provides stability, diversity, and long-range narrative richness.

4.2 Paging as Emergent Micro-Memory

Paging is the cognitive backbone of the architecture. Each agent maintains many *Token Pages*: small, local models that specialize in narrow transition contexts. A Token Page contains:

- a small set of input tokens,

- a small set of output tokens,
- a deterministic mapping for trivial transitions,
- or a tiny prediction network for multi-option transitions.

Token Pages are not designed; they *emerge* during training. New pages appear when new transition patterns arise; existing pages may split or refine as the curriculum expands. Paging therefore grows dynamically and reflects the statistical and structural regularities of the learned world rather than predefined semantic categories.

Pages often correlate with intuitive notions such as fissures, groves, or caverns, but this is not enforced. Interaction learning can blend or reshape pages, and each agent develops a unique paging topology—its cognitive fingerprint.

4.3 Dynamic and Incremental Learning

A DMLM begins as an empty system: no pages, no transitions, no canonical structures. All architecture emerges through learning. Because learning is local and modular:

- the model can generate coherent output at any stage of training,
- new examples can be introduced at any time,
- pages and transitions grow without global retraining,
- and the system supports lifelong, open-ended learning.

This dynamic property distinguishes DMLMs from transformers. A DMLM is not a frozen parameter block but a persistent, evolving architecture.

4.4 Transition Maps as Learned Continuity

Each agent maintains a sparse *transition map* that records which token-to-token transitions it has observed. Transition maps serve three roles:

- **Deterministic mappings** for highly stable transitions,
- **Bias-based N-mapping** during training to stabilize learning,
- **Logit filtering** during inference to softly suppress implausible transitions.

Transition maps complement paging by providing continuity between Token Pages. They encode the agent’s narrative highways—its preferred lemma clusters, verbs, and world physics—and become a core part of its personality.

4.5 Evaluators and Epistemic Governance

Neural components generate proposals; deterministic evaluators decide which proposals survive. DMLMs use several evaluators:

- **Grammar Engine:** deterministic rendering and syntactic validation,
- **Semantic Engine:** actor consistency, world physics, and semantic plausibility,
- **Coherence mechanisms:** rule-based and model-based checks for long-range stability,
- **Booster Engine:** controlled lexical, structural, and semantic variation.

Evaluators provide *epistemic governance*: they define the boundaries within which learning and generation remain stable. Strong evaluators produce strong emergent behavior.

4.6 The PRE Loop: Proposal–Retry–Evaluate

The PRE loop is the cognitive core of the architecture:

1. **Proposal:** an agent proposes a candidate sentence using its Token Pages, transition maps, and attention-compressed context.
2. **Retry:** if evaluation fails, the agent refines or regenerates the proposal.
3. **Evaluate:** deterministic evaluators accept or reject the sentence.

PRE transforms generation from single-shot sampling into iterative refinement. Transformers generate tokens without revision; Writer Agents generate sentences with revision cycles.

4.7 Canonical Form: Internal vs. External Language

DMLMs think in a canonical internal language and speak in natural language. Canonical form provides:

- lemma-level tokens,
- morphological operators,
- narrative-phase markers,
- and explicit control tokens.

The Grammar Engine deterministically renders canonical form into natural language. This strict separation ensures stability, interpretability, and perfect grammar even in tiny models.

4.8 Curriculum as Language Definition

The curriculum is not a dataset but a formal specification of the micro-language. It defines:

- grammar skeleton and structure tokens,
- semantic primitives and lemma clusters,
- narrative phases,
- attention patterns,
- and transition topology.

From a few dozen canonical sentences, agents develop grammar, semantics, roles, and world physics through interaction and evolution.

4.9 Learned Attention and Context Compression

Attention in DMLMs is learned relevance: a scalar importance score for each token in the context window. Attention is shaped by curriculum signals and social learning between agents.

Context compression filters out irrelevant tokens, increasing effective context size and simplifying prediction. This provides a lightweight, interpretable alternative to transformer attention.

Together, these components form a modular, dynamic, and extensible architecture for language generation. DMLMs replace monolithic parameterization with explicit cognitive structure, enabling stable long-form behavior, emergent worlds, and lifelong learning without transformers or large-scale training.

5 Relation to the Paged-PRE-CAS AGI Architecture

Dynamic Modular Language Models can be viewed as a concrete instantiation of the broader Paged-PRE-CAS architecture introduced in the Triadic Cosmos AGI framework. Although DMLMs operate at the level of sentence generation rather than token-level reasoning, the underlying structural principles are the same: paging for domain decomposition, PRE for iterative reasoning, and optional CAS vectors for epistemic guidance.

5.1 Paged Reasoning at the Level of Writer Agents

In the AGI architecture, Paging isolates reasoning into domain-specific pages that constrain the action space and activate domain-appropriate evaluators. In DMLMs, each Writer Agent implements the same principle through its Token Pages:

- each Token Page is a local reasoning context,
- deterministic or predictive transitions restrict the action space,
- cross-domain interference is structurally impossible,
- and each agent develops its own paging topology.

Thus, every Writer Agent is a *paged reasoning system*: a Paged-PRE unit specialized for narrative generation.

5.2 PRE: From Proposal–Evaluate–Refine to Proposal–Evaluate–Retry

In the AGI architecture, PRE is defined as:

Proposal → Evaluate → Refine

where the refinement step explicitly incorporates the previous proposal to produce an improved candidate. This yields a true recursive reasoning loop.

In DMLMs, PRE is implemented as:

Proposal → Evaluate → Retry

Retries do not yet reuse the previous proposal explicitly, but they operate within the same structural regime:

- retries select alternative transitions, lemma clusters, or Token Pages,
- evaluators provide deterministic feedback,
- the loop continues until a valid sentence is found.

This makes DMLM-PRE a *functional subset* of AGI-PRE: refinement is implicit rather than explicit. The architecture remains fully compatible with a future refine-based variant in which Writer Agents incorporate their own failed proposals as input for improved regeneration.

5.3 Optional Token-Level PRE for Efficiency

The AGI architecture applies PRE at the token level. DMLMs apply PRE at the sentence level, but token-level PRE remains an optional extension:

- token-level PRE can refine transitions inside a Token Page,
- it can reduce retries at the sentence level,
- and it can increase generation efficiency by filtering weak transitions earlier.

Token-level PRE is therefore not required for correctness, but it is a natural efficiency layer that can be added without altering the architecture.

5.4 Optional CAS Vectors for Epistemic Guidance

The AGI architecture introduces CAS (Content–Accuracy–Safety) vectors as orthogonal epistemic evaluators. DMLMs do not require CAS for stable narrative generation, but CAS vectors can be integrated seamlessly:

- CAS dimensions can act as additional evaluators,
- CAS-N ensembles can provide pluralistic scoring,
- CAS vectors can guide refinement or retries during PRE,
- and CAS can serve as a governance layer for safety-critical applications.

Thus, DMLMs are CAS-compatible without depending on CAS for core functionality.

5.5 M-Paged-PRE: A Multi-Agent Reasoning Ensemble

The AGI architecture allows multiple Paged-PRE-CAS systems to operate in parallel for robustness and plural reasoning. DMLMs instantiate this principle directly: the system is an ensemble of M Writer Agents, each a paged reasoning core with its own Token Pages, transition maps, and micro-networks.

The Writer Engine acts as an arbiter:

- rotating agents,
- switching after repeated failure,
- balancing stylistic and semantic diversity,
- and ensuring stable long-form generation.

This makes a DMLM an *M-Paged-PRE system*: a plural, structurally diverse ensemble of paged reasoning units.

5.6 Conceptual Alignment

The correspondence between the two architectures is direct:

AGI Architecture	DMLM Architecture
Paging	Token Pages per Writer Agent
PRE (Refine-based)	PRE (Retry-based, refine-compatible)
Token-level PRE	Optional efficiency layer
CAS vectors	Optional evaluator extension
Domain experts	Token Pages + transition maps
Arbiter	Writer Engine (agent selection)
Ensembles	Multiple Writer Agents

DMLMs therefore instantiate the same structural principles as the Paged-PRE-CAS AGI architecture, but specialized for narrative generation and optimized for small, modular micro-models.

6 Example Output

To illustrate how a Dynamic Modular Language Model behaves in practice, we present a short story generated by a prototype DMLM trained on a compact curriculum. The model consisted of domain-bound micro-models (fissure, ember, forest), each with its own paging structure and local attractors. Grammar, semantic, and coherence evaluators governed sentence acceptance through the PRE loop, and canonical form ensured consistent role and world-state representation.

The following fragment is reproduced verbatim from the companion Triadic Cosmos volume:

Pulse bursts past dripping lava surges along hollow ridge. The rabbit staggers within shifting fissure. Deep rumble shakes along stone spine. Branches loosen within shifting ember mist. The rabbit staggers along lashing mound. The rabbit breathes beside quivering ember mist. The rabbit breathes amid dim fissure. The rabbit breathes along dim molten shapes. Deep rumble vibrates along dim ember mist. The rabbit slips beside rippling ember mist. The raccoon crawls between lashing ember ledge. The rabbit slips along shifting soil. The rabbit staggers toward ridge. The fox halts past grasping branches. The fox stalks against silent fissure. The fox halts against faint ridge. The fox chases against silent tendril fissure. Within shifting dark the fox stalks fast along lessons learned tonight. The rabbit stumbles forward blocking grove. Forest surges closer with hungry intent.

This example illustrates how DMLMs generate structured, coherent text through modularity, paging, evaluators, and the PRE loop—without transformers, large datasets, or monolithic training.

7 Limitations and Open Questions

Dynamic Modular Language Models (DMLMs) introduce a new architectural paradigm, but the framework also raises several limitations and open questions that require further investigation.

7.1 Limitations

- **Prototype-scale implementation.** Current DMLM implementations operate at small scale using compact curricula and lightweight micro-models. While this demonstrates architectural viability, it remains unclear how the framework behaves under substantially larger domain sets or more complex reasoning tasks.

- **Absence of large-scale empirical benchmarks.** DMLMs have not yet been evaluated on standard NLP benchmarks or compared directly to transformer-based LLMs. The architecture is designed for modularity and structure rather than raw performance, but systematic evaluation remains an open task.
- **Manual curriculum design.** Micro-models currently rely on hand-crafted or semi-structured curricula. Automated curriculum generation, domain discovery, and self-expansion mechanisms are not yet implemented.
- **Evaluator design and tuning.** Evaluators play a central role in governing grammar, semantics, coherence, and world logic. Their design is currently heuristic and domain-specific. A more principled theory of evaluator construction and interaction is needed.
- **Lack of formal guarantees.** While DMLMs exhibit stable long-form behavior in practice, there are no formal proofs regarding convergence, stability, or bounded recursion under the PRE loop.

7.2 Open Questions

- **Scaling behavior.** How do DMLMs behave when the number of micro-models grows by orders of magnitude? Does modularity preserve coherence, or do new forms of interference emerge?
- **Automated domain formation.** Can a DMLM autonomously discover new domains, generate its own curricula, or split existing domains when complexity increases?
- **Learning dynamics.** What are the optimal training strategies for micro-models? Can evaluators themselves be learned or co-evolve with the system?
- **Hybrid architectures.** How might DMLMs integrate with or complement transformer-based models? Are hybrid systems—combining modular structure with neural embeddings—advantageous?
- **Theoretical characterization.** Can DMLMs be described in terms of computational graphs, dynamical systems, or formal grammars? What is the expressive power of a DMLM relative to classical LMs?

In summary, DMLMs provide a promising architectural alternative to monolithic language models, but their full capabilities, limitations, and theoretical properties remain open areas for future research.

8 Testable Predictions

Dynamic Modular Language Models (DMLMs) make several structural predictions that follow directly from their architecture. These predictions are falsifiable and can be evaluated using small-scale implementations, without requiring large datasets or transformer-based models. They concern stability, modularity, evaluator behavior, and the dynamics of the PRE loop.

8.1 Prediction 1: Stable Long-Range Coherence Through Paging

Because paging creates local memory and local attractors, a DMLM should maintain coherent long-range structure even when individual micro-models have no global context window. Narrative arcs, roles, and world-state variables should remain stable across hundreds of generation steps.

Falsification: If long-range coherence collapses or drifts uncontrollably despite correct paging, the architectural claim fails.

8.2 Prediction 2: Evaluator-Governed Convergence

Plural evaluators (grammar, semantics, coherence, world logic) should cause the PRE loop to converge toward stable outputs. Repeated Proposal–Retry–Evaluate cycles should reduce variance and increase structural consistency.

Falsification: If evaluators fail to reduce variance or if PRE cycles diverge, the evaluator governance hypothesis is incorrect.

8.3 Prediction 3: Domain-Bound Behavior of Micro-Models

Each micro-model is trained on a compact curriculum and bound to a specific domain. The prediction is that micro-models should remain within their domain even under pressure: they should not spontaneously drift into unrelated genres, topics, or styles.

Falsification: If micro-models consistently leak into other domains despite correct paging and evaluators, the domain-bound modularity claim is weakened.

8.4 Prediction 4: Emergent Structure Without Scale

DMLMs predict that grammatical structure, narrative roles, and world models can emerge from compact curricula without large-scale training. Small micro-models should exhibit structured behavior disproportionate to their size.

Falsification: If emergent structure only appears at large scale or with massive datasets, the architecture does not provide the claimed structural advantages.

8.5 Prediction 5: Reproducible Behavior Under Fixed Configuration

Given identical paging, evaluators, and micro-model parameters, a DMLM should produce highly reproducible outputs. Variation should be bounded and predictable, not chaotic.

Falsification: If repeated runs with identical settings produce inconsistent or unstable outputs, the deterministic control-flow hypothesis is challenged.

8.6 Prediction 6: Modular Extensibility Without Retraining

Adding a new domain-specific micro-model should extend the system’s capabilities without retraining existing modules. Existing domains should remain stable and unaffected.

Falsification: If adding a new module destabilizes existing domains or requires global retraining, the modular extensibility claim is invalid.

8.7 Prediction 7: PRE Loop Produces Distinct Regimes

The PRE loop should exhibit identifiable behavioral regimes: rapid convergence, slow convergence, oscillation, or evaluator-driven rejection. These regimes should be measurable and predictable across runs.

Falsification: If PRE behavior is random or unstructured, the loop does not function as a deterministic reasoning mechanism.

8.8 Prediction 8: Local World Models Are Persistent

Paging should allow micro-models to maintain local world-state variables (locations, roles, objects, intentions) across long sequences. These world models should persist even when other modules activate.

Falsification: If world-state variables collapse or reset unpredictably, the paging mechanism does not provide the claimed cognitive structure.

Together, these predictions provide a falsifiable empirical program for evaluating DMLMs. They do not depend on scale, data volume, or transformer architectures, but follow directly from the modular, evaluator-governed, paging-based design of the system.

8.9 Prediction 9: Generation Efficiency Improves with Better Evaluators and Curricula

Across multiple prototype implementations, DMLMs exhibit consistent structural behavior even when trained with different curricula, context lengths, or micro-model sizes. The primary quantity that varies is generation efficiency: the proportion of generated sentences that pass the grammar engine, semantic engine, and structural evaluators without requiring retries.

The architecture predicts that generation efficiency is not random but systematically improvable. Enhancing evaluator quality (e.g., more precise grammar rules, richer semantic constraints), refining curricula, or improving micro-model training should lead to higher generation efficiency without altering the underlying architecture. Because the PRE loop filters low-quality proposals, better evaluators should reduce the number of retries and increase the acceptance rate.

Generation efficiency varies widely depending on evaluator strictness, but the qualitative behavior of the system remains stable across models. This suggests that efficiency is a tunable performance metric rather than a structural limitation of the architecture.

Because the architecture is fully modular, DMLMs allow PRE mechanisms to operate at multiple levels of granularity. Earlier prototypes used token-level PRE loops, and combining these with sentence-level PRE may further increase generation efficiency. This kind of mechanical extensibility is not available in monolithic transformer architectures, where the generation mechanism is fixed.

Falsification: If evaluator improvements or curriculum refinements do not increase generation efficiency, or if efficiency remains unstable across models with similar settings, the hypothesis that efficiency reflects structural alignment between micro-models and evaluators would be weakened.

What Has Already Been Observed

Several of the predictions above are partially confirmed by existing prototype implementations. Small-scale DMLMs already exhibit stable long-range coherence through paging, evaluator-governed convergence under the PRE loop, and reproducible behavior under fixed configurations. These observations support the architectural claims that local memory, plural evaluators, and deterministic control flow can produce structured behavior without large-scale training.

Other predictions remain open but are strongly suggested by the compositional nature of the architecture. In particular, domain-bound modularity and modular extensibility have not yet been tested at scale, but the ability to train clusters of micro-models on separate domains and later compose them into a larger system follows directly from the agent-based design. The PRE loop provides a natural mechanism for selecting or coordinating agents based on evaluated performance, suggesting that larger DMLMs may self-organize through evaluator-driven composition.

These observations indicate that while DMLMs are at an early stage, the prototype already demonstrates several of the architectural behaviors predicted by the framework.

9 Conclusion

This paper introduced Dynamic Modular Language Models (DMLMs), a new architectural approach to language generation based on modular micro-models, explicit evaluators, paging structures, and a deterministic PRE loop. Unlike monolithic transformer-based LLMs, DMLMs

provide cognitive structure, local memory, and epistemic governance at the architectural level rather than relying on scale or implicit statistical correlations.

The prototypes developed so far demonstrate that small DMLMs can produce coherent long-form text, maintain stable world models, and exhibit reproducible behavior across runs. These observations suggest that modularity, evaluator-governed reasoning, and local paging offer a viable alternative to the monolithic design of current language models. At the same time, the architecture remains at an early stage: large-scale evaluation, automated curriculum generation, and principled evaluator design are open areas for future research.

The goal of this paper was not to present a mature production system, but to articulate the core mechanisms of a new model class and to provide a foundation for further exploration. The broader theoretical and conceptual context of DMLMs is developed in the *Triadic Cosmos* ecosystem, which offers a more complete account of the ontology, design principles, and empirical demonstrations underlying this work.

DMLMs open a path toward structured, interpretable, and extensible language models that do not depend on massive datasets or monolithic architectures. We hope this seed architecture encourages further experimentation, theoretical development, and integration with the wider research community.

A Triadic Cosmos Ecosystem Context

This paper is part of the *Triadic Cosmos* ecosystem, which develops a unified conceptual and computational framework spanning physical ontology and modular architectures for intelligent systems. All materials, extended notes, and evolving documentation are available in the public repository: <https://github.com/triadic-cosmos>.

References

- Anderson, John R. and Christian Lebiere (1998). “The Atomic Components of Thought”. In: *Lawrence Erlbaum Associates*.
- Baars, Bernard J. (1988). “A Cognitive Theory of Consciousness”. In: *Cambridge University Press*.
- Brown, Tom B. et al. (2020). “Language Models are Few-Shot Learners”. In: *Proceedings of NeurIPS*.
- Dehaene, Stanislas (2014). *Consciousness and the Brain: Deciphering How the Brain Codes Our Thoughts*. Viking Press.
- Fedus, William, Barret Zoph, and Noam Shazeer (2022). “Switch Transformers: Scaling to Trillion Parameter Models with Simple and Efficient Sparsity”. In: *Proceedings of JMLR*.
- Kaplan, Jared et al. (2020). “Scaling Laws for Neural Language Models”. In: *arXiv preprint arXiv:2001.08361*.
- Laird, John E. (2012). *The Soar Cognitive Architecture*. MIT Press.
- LangChain (2023). *LangChain: Building Applications with LLMs through Composability*. GitHub repository.
- Lepikhin, Dmitry et al. (2021). “GShard: Scaling Giant Models with Conditional Computation and Automatic Sharding”. In: *Proceedings of ICML*.
- Microsoft (2023). *Semantic Kernel: An SDK for Integrating AI with Programming Languages*. GitHub repository.
- Minsky, Marvin (1986). *The Society of Mind*. Simon and Schuster.
- Qiu, Lin et al. (2024). “Emergent Modularity in Large Language Models”. In: *Proceedings of ICML*.
- Radford, Alec, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever (2019). “Language Models are Unsupervised Multitask Learners”. In: *OpenAI Technical Report*.
- Research, Microsoft (2023). *AutoGen: Enabling Next-Gen Multi-Agent AI Applications*. GitHub repository.
- Shazeer, Noam et al. (2017). “Outrageously Large Neural Networks: The Sparsely-Gated Mixture-of-Experts Layer”. In: *Proceedings of ICLR*.
- Shen, Sheng et al. (2024). “ModuleFormer: Learning Modular Large Language Models”. In: *Proceedings of ICML*.
- Sun, Ron (2002). *Duality of the Mind: A Bottom-Up Approach Toward Cognition*. Lawrence Erlbaum Associates.
- Vaswani, Ashish et al. (2017). “Attention Is All You Need”. In: *Proceedings of NeurIPS*.
- Xiao, Zhengxuan et al. (2023). “Configurable Large Language Models”. In: *Proceedings of NeurIPS*.